

Introduction to Programming

SITAMS CSE (AI & ML) • R23 • I Year • I Semester • Course Code: 23ESC114T

Source basis: Official SITAMS R23 CSE (AI & ML) syllabus uploaded for this Study Hub. This concise revision guide preserves the syllabus topic statements and adds a study checklist; it is not a replacement for prescribed textbooks or faculty notes.

Unit 1: Introduction to Programming and Problem Solving

Syllabus topics

- History of Computers, Basic organization of a computer: ALU, input-output units, memory, program counter, Introduction to Programming Languages, Basics of a Computer Program- Algorithms, flowcharts
- Introduction to Compilation and Execution, Primitive Data Types, Variables, and Constants, Basic Input and Output, Operators and Expressions, Type Conversion, and Casting
- Problem solving techniques: Algorithmic approach, characteristics of algorithm, Problem solving strategies: Top-down approach, Bottom-up approach, Time and space complexities of algorithms

Revision checklist

- Know the definitions and principles named in the syllabus.
- Prepare the standard methods, diagrams, formulas or algorithms relevant to the listed topics.
- Practice representative examples or problems from the prescribed study material.
- Review this unit against the official SITAMS syllabus before attempting its quiz.

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Unit 2: Control Structures

Syllabus topics

- Simple sequential programs, Conditional Statements (if, if-else, switch), Loops (for, while, do- while)
- Break and Continue

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Unit 3: Arrays and Strings

Syllabus topics

- Arrays indexing, memory model, programs with array of integers, two dimensional arrays
- Strings-Declaring and Initializing String Variables, Reading string from terminal, Writing string to the screen, String Handling Functions

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Unit 4: Pointers & User Defined Data types

Syllabus topics

- Pointers, dereferencing and address operators, pointer and address arithmetic, array manipulation using pointers
- User-defined data types-Structures and Unions

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Unit 5: Functions & File Handling

Syllabus topics

- Introduction to Functions, Function Declaration and Definition, Function call Return Types and Arguments, modifying parameters inside functions using pointers, arrays as parameters
- Scope and Lifetime of Variables
- File Handling- Basic Operations on Files – File Handling Function

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Study Hub status

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